

Machine Learning Handbook

What is Machine Learning?

Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation. Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation. Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation.

Types of Machine Learning

Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties. Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties. Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties.

Data Preprocessing

Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets. Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets. Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets.

Feature Engineering

Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization. Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization. Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization.

Common Algorithms

Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector

Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns. Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns. Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns.

Model Evaluation

Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance. Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance. Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance.

Overfitting and Underfitting

Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization. Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization. Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization.

Natural Language Processing

NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering. NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering. NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering.

Embeddings and Vector Databases

Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems. Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems. Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems.

Applications

Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education. Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education. Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education.

Best Practices

Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models. Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models. Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models.

What is Machine Learning?

Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation. Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation. Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation.

Types of Machine Learning

Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties. Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties. Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties.

Data Preprocessing

Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets. Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into

training and testing sets. Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets.

Feature Engineering

Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization. Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization. Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization.

Common Algorithms

Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns. Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns. Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns.

Model Evaluation

Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance. Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance. Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance.

Overfitting and Underfitting

Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization. Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization. Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization.

Natural Language Processing

NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering. NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering. NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering.

Embeddings and Vector Databases

Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems. Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems. Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems.

Applications

Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education. Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education. Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education.

Best Practices

Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models. Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models. Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models.

What is Machine Learning?

Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation. Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation. Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation.

Types of Machine Learning

Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties. Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties. Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties.

Data Preprocessing

Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets. Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets. Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets.

Feature Engineering

Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization. Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization. Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization.

Common Algorithms

Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns. Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns. Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns.

Model Evaluation

Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance. Regression models are evaluated using MAE, MSE,

RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance. Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance.

Overfitting and Underfitting

Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization. Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization. Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization.

Natural Language Processing

NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering. NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering. NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering.

Embeddings and Vector Databases

Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems. Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems. Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems.

Applications

Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education. Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education. Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education.

Best Practices

Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters,

document experiments, and monitor deployed models. Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models. Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models.

What is Machine Learning?

Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation. Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation. Machine Learning (ML) is a field of Artificial Intelligence that enables computers to learn patterns from data instead of relying only on explicitly programmed rules. Models improve as they are exposed to more data and are widely used for prediction, classification, recommendation, and automation.

Types of Machine Learning

Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties. Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties. Supervised learning uses labeled data to predict outputs. Unsupervised learning discovers hidden patterns in unlabeled data. Semi-supervised learning combines small amounts of labeled data with large amounts of unlabeled data. Reinforcement learning trains an agent through rewards and penalties.

Data Preprocessing

Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets. Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets. Real-world data often contains missing values, duplicate records, inconsistent formatting, and outliers. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features, feature engineering, and splitting the dataset into training and testing sets.

Feature Engineering

Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization. Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization. Feature engineering improves model performance by creating meaningful variables from existing data. Examples include extracting dates, creating ratios, polynomial features, interaction features, binning continuous variables, and text vectorization.

Common Algorithms

Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns. Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns. Linear Regression predicts continuous values. Logistic Regression solves binary classification. Decision Trees create rule-based models. Random Forest combines multiple trees. Support Vector Machines maximize the decision boundary. K-Nearest Neighbors classifies based on nearby samples. Naive Bayes is based on probability. Neural Networks learn complex patterns.

Model Evaluation

Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance. Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance. Regression models are evaluated using MAE, MSE, RMSE, and R-squared. Classification models use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. Cross-validation helps estimate generalization performance.

Overfitting and Underfitting

Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization. Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization. Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture the underlying pattern. Techniques like regularization, pruning, cross-validation, and more data help improve generalization.

Natural Language Processing

NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering. NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering. NLP enables computers to understand human language. Common tasks include tokenization, stopword removal, stemming, lemmatization, embeddings, sentiment analysis, named entity recognition, machine translation, and question answering.

Embeddings and Vector Databases

Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems. Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems.

Sentence embeddings convert text into dense numerical vectors that capture semantic meaning. Vector databases such as FAISS efficiently store these vectors and perform similarity search, making them essential for semantic search and Retrieval-Augmented Generation systems.

Applications

Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education. Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education. Machine learning powers recommendation systems, fraud detection, healthcare diagnostics, autonomous vehicles, predictive maintenance, finance, cybersecurity, chatbots, document search, image recognition, speech recognition, and personalized education.

Best Practices

Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models. Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models. Always understand the business problem, explore the dataset thoroughly, preprocess data carefully, choose appropriate evaluation metrics, compare multiple models, tune hyperparameters, document experiments, and monitor deployed models.